Year 9 Metal — Program and Scope & Sequence

Teacher-developed, source-backed review draft · current course evidence with 2025 syllabus forward alignment

Stage 5 · Industrial Technology – Metal | **Local review set v1.0** | **Prepared 5 August 2026**

Status: Local review document only. It is not NESA approved and does not change the live course, formal notifications or school program. Current local decisions and unavailable evidence remain Teacher to confirm.

Source authority

Source role	Exact basis used
Current course source	Local Yr-9-Metal checkout at commit 6e1249ec090d459d0dc4d07d9379a9e1a05e8f4c; live canonical course evidence checked by TASK 12A.
Course evidence	guided/data.js; COURSE-CONTRACT.md; SOURCE-NOTES.txt; PLAN-SOURCE-MANIFEST.md; folio and 2026 task/exam pages.
Authoritative plans	TOOL CARRYALL DEVELOPMENT and FISHING ROD HOLDER: COMPONENTS, copied unchanged and source-manifested in the current course.
Assessment authority	Current course records Task 1 T2 W7 40%, Task 3 exam T4 W5 20%, Task 2 T4 W6 40%; final handbook/notification controls.
Current statutory position	Industrial Technology Years 7–10 Syllabus (2019) remains the current authority unless local early implementation is confirmed.
Forward-alignment authority	NSW Industrial Technology 7–10 Syllabus (2025): implementation from 2028; 2026–2027 are planning/preparation years. Official outcome/content pages and official Metal sample scope were checked.
Benchmark only	current-stage-5-timber-program.docx controls broad program structure and visual rhythm only; no Timber content or values were copied.

Whole-year scope overview

Term	Project/unit	Learning and evidence emphasis	Status
Term 1	Sheet-metal Toolbox foundations and controlled production	Safety; plans; materials; marking; cutting/forming; quality evidence	Indicative placement from current source — exact pacing Teacher to confirm
Term 2	Toolbox completion and Project 2 transition	Joining/assembly; finishing; evaluation; Task 1 T2 W7; Fishing Rod Holder brief/plan	Assessment facts current; remaining sequence Teacher to confirm
Term 3	Fishing Rod Holder development and production	Plan reading; planning; component production; assembly; collaboration opportunity	Exact processes and collaboration Teacher to confirm
Term 4	Testing, industry context and formal assessment	Evaluation; material life cycle; careers/enterprise; exam T4 W5; Task 2 T4 W6	Cultural/industry evidence and W7–10 Teacher to confirm

Current guided-course spine

Module	Current weeks	Title	Practical/theory focus	Evidence or project position
1	Weeks 1–2	Workshop systems and risk management	Hazards, risk, hierarchy of controls, safe practice and incident reporting.	Risk response; checks; teacher observation.
2	Weeks 3–4	Metals, project briefs and planning	Properties/performance, two briefs, design planning and verified plan reading.	Material justification; 20 plan-reading checks; annotated evidence.

3	Weeks 5–6	Mark, cut and form accurately	Datum edges, waste-side marking, workholding, cutting and forming sequence.	Accuracy/process explanation; practical checks.
4	Weeks 7–8	Join, machine and finish with purpose	Joining selection, trial fit, permission-based machining and protective finishes.	Joining justification; practical and finish evidence.
5	Weeks 9–10	Technology, quality and evaluation	Conventional/CNC suitability, quality evidence, folio and evaluation.	Technology comparison and evidence-based evaluation.

Detailed teaching sequence

Sequencing status: The content and evidence are source-grounded. Term/week placement is an editable planning allocation unless a current assessment week is stated. Exact lesson hours, interruptions and practical pace are Teacher to confirm.

Term 1 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T1 W1–2	Workshop induction; hazards, risk, hierarchy of controls, WHS documents, first aid and incident response.	Apply risk assessment, permission and stop/reassess decisions to every practical stage.	Safety induction; plan-linked risk response; teacher observation; folio WHS evidence.	SAF, COM, DES	Local PPE, SOPs and emergency procedures — Teacher to confirm
T1 W3–4	Toolbox brief, requirements and constraints; read the verified Tool Carryall development drawing.	Locate plan facts before marking material; record and query anything unclear instead of guessing.	Brief/specification; plan-reading checks; annotated plan or folio note.	COM, DES, SAF	Final brief, dimensions and drawing expectations — Teacher to confirm
T1 W5–6	Ferrous/nonferrous metals; forms, properties, performance, resource planning and introductory cost thinking.	Justify the teacher-specified metal/stock form against function, fabrication, corrosion, availability and finish.	Materials investigation; justification; draft material/cutting list.	USE, DES, ENV	Actual materials, quantities, prices and costing — Teacher to confirm
T1 W7–8	Production planning; datum/reference edge; accurate measuring and marking; waste side; workholding; checkpoints.	Use the approved production plan and teacher demonstration to control accuracy before irreversible work.	Production plan; drawing evidence; marking/measurement observation; checkpoint record.	DES, COM, USE, SAF, PRO	Exact sequence and student-created CAD/drawing requirement — Teacher to confirm
T1 W9–10	Teacher-approved cutting and forming; tool/equipment selection; safe setup; material conservation.	Connect material properties and geometry to the demonstrated process; inspect before and after each operation.	Practical observation; staged production evidence; defect and quality checks.	SAF, USE, INP, PRO, ENV	Actual tools/processes, supervision and waste routine — Teacher to confirm

Term 2 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T2 W1–3	Teacher-approved joining, machining/assembly, trial fit, surface preparation and finish.	Justify joining using load, permanence, access, heat effects, appearance and corrosion; trial-fit before permanent work.	Joining decision; trial-fit record; practical observation; finish decision.	SAF, USE, INP, PRO, EVL	Joining method, specialist process and finish system — Teacher to confirm
T2 W4–6	Quality assurance/control; functional and construction checks; corrective action; evaluation; report/folio completion.	Compare the product with the approved brief and use specific measurement, fit, function and process evidence.	Completed Toolbox; test/quality record; problem-solving account; final evaluation.	EVL, DES, COM, PRO, INP	Test and marking criteria; submission workflow — Teacher to confirm
T2 W7	Formal Task 1: Practical Project & Report 1 — Sheet-metal Toolbox (40%).	Assessment follows the current handbook and approved teacher-issued notification.	Completed practical project and authorised evidence package.	Approved task outcomes	Exact date/time, rubric, marks and submission — Teacher to confirm
T2 W8–10	Transfer Toolbox learning to Fishing Rod Holder; introduce second brief, drawing, environment, parts and evidence.	Identify transferable skills and project-specific differences; never copy unsupported Toolbox specifications.	Project 1 reflection; Project 2 brief; plan-reading record; initial risk/material analysis.	COM, DES, USE, SAF, EVL	Transition timing and separate/reused folio workflow — Teacher to confirm

Term 3 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap

T3 W1–2	Safety refresh; read Fishing Rod Holder components drawing; identify holder, stake and rotation tab.	Link function and outdoor exposure to the teacher-specified material/profile and safe fabrication.	Plan-reading checks; risk record; materials justification.	SAF, COM, USE, DES, ENV	Exact dimensions, materials, processes and finish — Teacher to confirm
T3 W3–4	Graphical communication; production plan; materials, quantities and costs; collaboration roles; quality plan.	Use written and graphical evidence to control the approved fabrication sequence.	Sketch/drawing/CAD; material/cost list; role allocation; quality plan.	COM, DES, USE, SAF	Graphical form, prices, collaboration and quality format — Teacher to confirm
T3 W5–7	Teacher-approved marking, cutting, shaping, forming, machining and detailing; precision measurement and maintenance.	Analyse why each tool/process suits the component and check against the issued drawing or template.	Practical observation; process log; measurement/check record; setup explanation.	SAF, USE, INP, PRO, COM	Process range, lathe/thread/heat status and SOPs — Teacher to confirm
T3 W8–10	Teacher-approved joining, assembly, trial fit and finish; planned collaborative production opportunity.	Compare options and apply the approved method; record each student's contribution to a shared task.	Joining/finish justification; assembly evidence; collaboration record.	SAF, USE, INP, PRO, EVL, IND	Joining/finish method and collaboration evidence — Teacher to confirm

Term 4 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T4 W1–2	Functional testing; tolerance, alignment and fit; defect analysis; controlled correction; four-quality evaluation.	Use authorised tests and objective criteria, not appearance alone.	Test/quality record; correction case study; final folio evidence.	EVL, INP, PRO, DES, SAF	Testing criteria, tolerances and economic/environmental evidence — Teacher to confirm
T4 W3–4	Metal industry in context: ores/processes, ethical sourcing, Country/Community impacts, recycling, careers and enterprise.	Use both projects as material-life-cycle and manufacturing case studies; follow authorised Cultural protocols.	Impact investigation; career/enterprise profile; manufacturing comparison; exam preparation.	ENV, IND, COM, USE	Cultural authority, industry link and career task — Teacher to confirm
T4 W5	Formal Task 3: Final Examination (20%).	Complete under approved assessment conditions; written evidence does not replace practical observation.	Submitted examination response.	Approved task outcomes	Exact date, conditions and 2025 blueprint — Teacher to confirm
T4 W6	Formal Task 2: Project & Report 2 — Fishing Rod Holder (40%).	Assessment follows the current handbook and approved teacher-issued notification.	Completed practical project and authorised evidence package.	Approved task outcomes	Exact date/time, rubric, marks and submission — Teacher to confirm
T4 W7–10	Post-assessment correction, extension, evidence moderation, maintenance and course evaluation.	Use only if authorised after formal assessment completion.	Targeted remediation/extension; student/program evaluation.	Teacher to confirm	Whether this time is available and its purpose — Teacher to confirm

Outcome and evidence map

Implementation note: Formal 2026 assessment pages currently use the 2019 IND5 outcome bank. The INT5 mapping below is a forward-planning layer for the 2025 syllabus and must not replace formal codes until local implementation is authorised.

2025 outcome	Official wording	Coverage	Current evidence and required action
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts	Strong	Module 1, risk response, WHS folio, teacher observation, practical projects and exam.
INT5-COM-01	communicates ideas and processes in industrial technology contexts	Strong/partial	Plan reading, written responses and folio; student-created graphics/CAD and safe data management need confirmation.
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment	Partial	Material/finish and technology prompts; ores, Country/Community impacts, ethical sourcing and recycling need explicit evidence.
INT5-DES-01	selects and applies design, project management and technical skills in project production	Strong	Briefs, constraints, planning, cutting/resources, checkpoints, problem solving and evaluation.
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects	Strong	Material justification, tool/process selection and teacher-approved project decisions.

INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products	Strong/partial	Quality, testing, problem solving and evaluation; economic/environmental criteria need strengthening.
INT5-INP-01	analyses and applies a range of industrial processes in production contexts	Strong opportunity	Marking, cutting, forming, joining, machining and finishing; exact practical/demonstration status is teacher-controlled.
INT5-PRO-01	demonstrates practical skills in the production of projects	Strong if recorded	Two practical projects and staged production; achievement requires direct observation evidence.
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices	Weak/partial	Current/emerging manufacturing entry points; careers and enterprise evidence must be added.

Formal assessment and evidence register

Task	Current title/context	Timing	Weight	Evidence role	Teacher-controlled fields
Task 1	Practical Project & Report 1 — Sheet-metal Toolbox	Term 2 Week 7	40%	Project, direct practical observation, report/folio, quality/evaluation	Exact date/time, outcomes, rubric, marks and submission — Teacher to confirm
Task 3	Final Examination	Term 4 Week 5	20%	60-minute digital/written evidence across course knowledge and application	Exact date, conditions, outcome blueprint and approval — Teacher to confirm
Task 2	Project & Report 2 — Fishing Rod Holder	Term 4 Week 6	40%	Project, direct practical observation, report/folio, testing and evaluation	Exact date/time, outcomes, rubric, marks and submission — Teacher to confirm

Teacher to confirm register

No.	Decision/evidence required	Status
1	Whether this is a standalone 100-hour course or one year of a 200-hour Stage 5 Metal elective.	Teacher to confirm
2	When the school/sector will adopt the 2025 syllabus and the formal reporting-code transition.	Teacher to confirm
3	Exact lesson allocation, calendar weeks, interruptions and practical pace.	Teacher to confirm
4	Current task notices: exact dates/times, outcomes, marks/rubrics, submission, approval and absence/misadventure details.	Teacher to confirm
5	Final briefs, drawings, materials, dimensions, tolerances, hardware, sequence, testing criteria and finishes.	Teacher to confirm
6	Local WHS: PPE, SOPs, SDSs, machinery access, supervision, risk assessments, settings and emergency procedures.	Teacher to confirm
7	Which specialist processes are practical, demonstrated, simulated/investigated or theory-only.	Teacher to confirm
8	The required collaborative activity, roles, common output and evidence of individual contribution.	Teacher to confirm
9	Authorised Aboriginal and Torres Strait Islander knowledge sources, Community protocols and ICIP boundaries.	Teacher to confirm
10	Required student-created graphical communication and applicable drawing/CAD expectations.	Teacher to confirm
11	Quantities/costing evidence, safe data-management practice and actual submission workflow.	Teacher to confirm
12	Careers/enterprise/industry evidence and the current source or industry link used.	Teacher to confirm
13	Accessibility adjustments and any separate Life Skills programming decisions.	Teacher to confirm
14	How the Year 9 modules are distributed/revisited across the two project semesters.	Teacher to confirm
15	Authorised use of Term 4 Weeks 7–10.	Teacher to confirm

Program evaluation and approval

Field	Entry
Teacher/program author	Teacher to confirm
Faculty review date	Teacher to confirm
Syllabus implementation basis	Teacher to confirm
Evaluation notes and revisions	
Approval/version	Teacher to confirm